The second point follows immediately from the definition of $\pi_\gamma p_\theta$. For the third point, we can estimate that

$$\begin{aligned}
\|\pi_\gamma p_\theta(x) - p_{\theta^\star}(x)\| &= \left\| \frac{p_\theta(x) + \gamma p_{\theta^\star}(x)}{1+\gamma} - \frac{p_{\theta^\star}(x)(1+\gamma)}{1+\gamma} \right\| \\
&= \frac{1}{1+\gamma} \cdot \|p_\theta(x) - p_{\theta^\star}(x)\| \\
&\leq \frac{\gamma}{1+\gamma} \cdot \|p_{\theta^\star}(x) - p_\theta(x)\| \\
&= \left\| \frac{p_\theta(x) + \gamma p_{\theta^\star}(x)}{1+\gamma} - \frac{p_\theta(x)(1+\gamma)}{1+\gamma} \right\| \\
&= \|\pi_\gamma p_\theta(x) - p_\theta(x)\|
\end{aligned}$$

when $\gamma > 1$. The inequality flips when $\gamma < 1$. □

Intuitively, it is clear that we cannot hope to prove general results about generative models without assuming something about the mapping $\theta \mapsto p_\theta$. We now state the two assumptions we require in order to make our theoretical arguments; note that they are precisely the same assumptions made in (Bertrand et al., 2024). The first assumption is a local Lipschitzness property that we will exploit via the Kantorovich-Rubenstein duality:

Assumption A.3. For θ close enough to $\theta^\star$, the mapping $x \mapsto \nabla_\theta \nabla_\theta \log p_\theta(x)$ is L-Lipschitz.

The second assumption is a local regularity and concavity condition:

Assumption A.4. The mapping $\theta \mapsto \mathbb{E}_{x\sim p_{\text{data}}}[\log p_\theta(x)]$ is continuously twice differentiable locally around $\theta^\star$ and $\mathbb{E}_{x\sim p_{\text{data}}}\left[\nabla_\theta \nabla_\theta \log p_\theta(x)\right]_{\theta^\star} \preceq -\alpha I_d \prec 0$.

We next show the existence and uniqueness of $\pi_\gamma \mathcal{G}_\lambda^\infty(\infty)$ locally around $\theta^\star$.

Proposition A.5 (The Local Maximum Likelihood Solution is Unique)**.** *The following are true:*

A. *There exists an open neighborhood $U \subset \mathbb{R}^d$ containing $\theta^\star$ and a continuous function $g : U \to \mathbb{R}^d$ such that $g(\theta^\star) = \theta^\star$, and*

$$\nabla_{\theta'} \mathcal{H}(\theta, \theta')|_{\theta, g(\theta)} = 0 \tag{14}$$

for every $\theta \in U$.

B. *Given optimal model parameters $\theta^\star$ as in (9) that follow Assumptions A.3 and A.4, we have that, if $\varepsilon L < \alpha$, then for all $\lambda > 0$ and θ in a small enough neighborhood U around $\theta^\star$, there exists a unique local maximizer $\pi_\gamma \mathcal{G}_\lambda^\infty(\theta)$ in U.*

Proof. **We first prove part A.** It suffices to apply the Implicit Function Theorem to the map

$$\mathbb{R}^{2d} \to \mathbb{R}^d : (\theta, \theta') \mapsto \nabla_{\theta'} \mathcal{H}(\theta, \theta')|_{\theta, \theta'} \tag{15}$$

in an open neighborhood of $(\theta^\star, \theta^\star)$. To do this, we need to show the following:

i) The map vanishes at $(\theta^\star, \theta^\star)$, i.e.

$$\nabla_{\theta'} \mathcal{H}(\theta, \theta')|_{\theta^\star, \theta^\star} = 0. \tag{16}$$

ii) The Jacobian matrix at $(\theta^\star, \theta^\star)$ is invertible, i.e.,

$$\nabla_{\theta'} \nabla_{\theta'} \mathcal{H}(\theta, \theta')|_{\theta^\star, \theta^\star} \qquad \text{is invertible.} \tag{17}$$

We first prove i). Recall from the definition (10) that $\pi_\gamma \mathcal{G}_\lambda^\infty(\theta) = \arg\max_{\theta' \in \Theta} \mathcal{H}(\theta, \theta')$. This means that for any θ, $\pi_\gamma \mathcal{G}_\lambda^\infty(\theta)$ is the choice of θ' which maximizes $\mathcal{H}(\theta, \theta')$. In particular, for $\theta = \theta^\star$, we have that $\theta' = \pi_\gamma \mathcal{G}_\lambda^\infty(\theta^\star)$ is the choice which maximizes $\mathcal{H}(\theta^\star, \theta')$. But $\pi_\gamma \mathcal{G}_\lambda^\infty(\theta^\star) = \theta^\star$ by Proposition A.6. This implies that its derivative is zero at $\theta' = \theta^\star$, meaning $\nabla_{\theta'} \mathcal{H}(\theta, \theta')|_{\theta^\star, \theta^\star} = 0$, as needed.

Now we prove ii). In order to show that the matrix (17) is invertible, it suffices to show it is close to another matrix which is invertible. A natural choice is the matrix

$$M = (1+\lambda)\nabla_{\theta'}\nabla_{\theta'}\mathbb{E}_{x\sim p_{\text{data}}}[\log p_{\theta'}(x)]|_{\theta^\star}. \tag{18}$$

First of all, note that this matrix indeed exists; by Assumption 2 A.4, we know the map $\theta' \mapsto \mathbb{E}_{x\sim p_{\text{data}}}[\log p_{\theta'}(x)]$ is continuously twice differentiable locally near $\theta^\star$. We can estimate that the matrices (17) and (18) are indeed close as follows:

$$\begin{aligned}
&\|\nabla_{\theta'}\nabla_{\theta'}\mathcal{H}(\theta,\theta')|_{\theta^\star,\theta^\star} - (1+\lambda)\nabla_{\theta'}\nabla_{\theta'}\mathbb{E}_{x\sim p_{\text{data}}}[\log p_{\theta'}(x)]_{\theta^\star}\| \\
&\quad = \|\nabla_{\theta'}\nabla_{\theta'}[\mathbb{E}_{x\sim p_{\text{data}}}\log p_{\theta'}(x) + \lambda\mathbb{E}_{x\sim\pi_\gamma p_\theta} p_{\theta'}(x)]|_{\theta^\star,\theta^\star} - (1+\lambda)\nabla_{\theta'}\nabla_{\theta'}\mathbb{E}_{x\sim p_{\text{data}}}[\log p_{\theta'}(x)]_{\theta^\star}\| \\
&\quad = \lambda\|[\nabla_{\theta'}\nabla_{\theta'}\mathbb{E}_{x\sim\pi_\gamma p_\theta}\log p_{\theta'}(x)]|_{\theta^\star,\theta^\star} - \nabla_{\theta'}\nabla_{\theta'}\mathbb{E}_{x\sim p_{\text{data}}}[\log p_{\theta'}(x)]_{\theta^\star}\| \\
&\quad = \lambda\|[\nabla_{\theta'}\nabla_{\theta'}\mathbb{E}_{x\sim p_{\theta^\star}}\log p_{\theta'}(x)]_{\theta^\star} - [\nabla_{\theta'}\nabla_{\theta'}\mathbb{E}_{x\sim p_{\text{data}}}\log p_{\theta'}(x)]_{\theta^\star}\| \\
&\quad = \lambda\|[\mathbb{E}_{x\sim p_{\theta^\star}}\nabla_{\theta'}\nabla_{\theta'}\log p_{\theta'}(x)]_{\theta^\star} - [\mathbb{E}_{x\sim p_{\text{data}}}\nabla_{\theta'}\nabla_{\theta'}\log p_{\theta'}(x)]_{\theta^\star}\| \\
&\quad \le L\lambda\mathbb{E}_{(x,x')\sim p_{\theta^\star}\times p_{\text{data}}}\|[\nabla_{\theta'}\nabla_{\theta'}\log p_{\theta'}(x)]_{\theta^\star} - \nabla_{\theta'}\nabla_{\theta'}\log p_{\theta'}(x)]_{\theta^\star}\| \\
&\quad \le \lambda\varepsilon L
\end{aligned}$$

where the first equality follows from the definition of $\mathcal{H}$ in (13); the second equality follows from some cancellation; the third equality follows the fact that the derivatives are constant with respect to θ, and $\pi_\gamma p_{\theta^\star} = p_{\theta^\star}$ by Lemma A.2; we exchange the derivative and the expectation in equation 4 using the Dominated Convergence Theorem, since Assumption 1 A.3 says that $x \mapsto \nabla_\theta\nabla_\theta \log p_\theta(x)$ is L-Lipschitz; the fifth estimate follows from Kantorovich-Rubinstein Duality; and the final estimate is the definition of Wasserstein distance (12).

Finally, we verify M is indeed invertible. Assumption 2 A.4 implies that the largest eigenvalue of M is at most $-(1+\lambda)\alpha$. Therefore, since all eigenvalues of M are nonzero, M is invertible. We can now apply the implicit function theorem to (15), and part A follows immediately.

Next, we prove part B. Let $d_U = \sup_{\theta\in U} d_W(p_{\theta^\star}, p_\theta)$. To verify that $g(\theta)$ is a local maximizer of (15), it suffices to show that $\nabla_{\theta'}\nabla_{\theta'}\mathcal{H}(\theta, g(\theta)) \prec 0$. By Assumption 2 A.4, we know $\nabla_{\theta'}\nabla_{\theta'}\mathcal{H}_1(\theta^\star) \prec -\alpha I_d$ and since $\theta' \mapsto \nabla_{\theta'}\nabla_{\theta'}\mathcal{H}_1(\theta')$ is *continuously* twice differentiable locally near $\theta^\star$, we also have $\nabla_{\theta'}\nabla_{\theta'}\mathcal{H}_1(g(\theta)) \prec -\alpha I_d$. Thus, we have

$$\begin{aligned}
\nabla_{\theta'}\nabla_{\theta'}\mathcal{H}(\theta, g(\theta)) &= \nabla_{\theta'}\nabla_{\theta'}\mathcal{H}_1(g(\theta')) + \lambda\nabla_{\theta'}\nabla_{\theta'}\mathcal{H}_2(\theta, g(\theta)) \\
&= (1+\lambda)\nabla_{\theta'}\nabla_{\theta'}\mathcal{H}_1(g(\theta)) + \lambda(\nabla_{\theta'}\nabla_{\theta'}\mathcal{H}_2(\theta, g(\theta)) - \nabla_{\theta'}\nabla_{\theta'}\mathcal{H}_1(g(\theta))) \\
&\preceq -\alpha(1+\lambda)I_d + \lambda L\left(\frac{1}{1+\gamma}d_W(p_\theta, p_{\theta^\star}) + \varepsilon\right)I_d,
\end{aligned}$$

where the last step follows from Kantorovich-Rubsenstein duality:

$$\begin{aligned}
&\|\nabla_{\theta'}\nabla_{\theta'}\mathcal{H}_2(\theta,\theta') - \nabla_{\theta'}\nabla_{\theta'}\mathcal{H}_1(\theta')\| \\
&\quad \le \|\nabla_{\theta'}\nabla_{\theta'}\mathcal{H}_2(\theta,\theta') - \nabla_{\theta'}\nabla_{\theta'}\mathcal{H}_2(\theta^\star,\theta')\| + \|\nabla_{\theta'}\nabla_{\theta'}\mathcal{H}_2(\theta^\star,\theta') - \nabla_{\theta'}\nabla_{\theta'}\mathcal{H}_1(\theta')\| \\
&\quad = \|\int_{\mathbb{R}^d}\nabla_{\theta'}\nabla_{\theta'}\log p_{\theta'}(x)\frac{p_\theta(x)+\gamma p_{\theta^\star}(x)}{1+\gamma}\,dx - \int_{\mathbb{R}^d}\nabla_{\theta'}\nabla_{\theta'}\log p_{\theta'}(x)p_{\theta^\star}(x)\,dx\| \\
&\qquad + \|\mathbb{E}_{x\sim p_{\text{data}}}[\log p_{\theta'}(x)] - \mathbb{E}_{x\sim p_{\theta^\star}}[\log p_{\theta'}(x)]\| \\
&\quad \le \frac{1}{1+\gamma}\|\int_{\mathbb{R}^d}\nabla_{\theta'}\nabla_{\theta'}\log p_{\theta'}(x)\left(p_\theta(x) - p_{\theta^\star}(x)\right)\,dx\| + L\varepsilon \\
&\quad = \frac{1}{1+\gamma}\|\mathbb{E}_{x\sim p_\theta}[\log p_{\theta'}(x)] - \mathbb{E}_{x\sim p_{\theta^\star}}[\log p_{\theta'}(x)]\| + L\varepsilon \\
&\quad \le \frac{L}{1+\gamma}d_W(p_\theta, p_{\theta^\star}) + L\varepsilon \\
&\quad \le \frac{L}{1+\gamma}d_U + L\varepsilon
\end{aligned}$$

Thus, to have $\nabla_{\theta'}\nabla_{\theta'}\mathcal{H}(\theta, g(\theta)) \prec 0$, it is sufficient that

$$-\alpha(1+\lambda) + \lambda L\left(\frac{1}{1+\gamma}d_U + \varepsilon\right) < 0,$$

which is guaranteed for all $\lambda > 0$ by $\alpha > L\varepsilon$ and $d_U \leq \frac{\alpha(1+\gamma)}{\lambda}$. This concludes the proof. □

Further, as we would expect, $\theta^\star$ is a fixed point of $\pi_\gamma \mathcal{G}_\lambda^\infty$:

Proposition A.6 (The optimal parametric generative model is a fixed point)**.** *For any given data distribution p_{data}, any $\theta^\star$ as defined by (9), and for all $\lambda > 0$, we have $\pi_\gamma \mathcal{G}_\lambda^\infty(\theta^\star) = \theta^\star$.*

Proof. Unpacking definition (10) shows that $\pi_\gamma \mathcal{G}_\lambda^\infty(\theta^\star) = \mathcal{G}_\lambda^\infty(\theta^\star)$, and we know by Proposition 4 from (Bertrand et al., 2024) that $\mathcal{G}_\lambda^\infty(\theta^\star) = \theta^\star$. □

A.2. Convergence of Iterative Fine-tuning with Correction for Infinite Sampling

We now have the required setup to state and prove a convergence result for iterative fine-tuning assuming infinite access to underlying probablity distributions. We need the following result, which is a technical lemma that provides a computation of the Jacobian of $\pi_\gamma G_\lambda^\infty$ at $\theta^\star$ as well as a spectral bound, both essential for the proof of Theorem A.8.

Lemma A.7. *We define the matrices*

$$A := (\nabla^2_{\theta',\theta'} \mathcal{H}_1(\theta'))|_{\theta^\star} \tag{19}$$

$$B := \nabla^2_{\theta,\theta'} \mathbb{E}_{x\sim p_\theta}[\log p_{\theta'}(x)]\big|_{\theta^\star,\theta^\star} \tag{20}$$

$$C := \nabla^2_{\theta',\theta'} \mathbb{E}_{x\sim p_\theta}[\log p_{\theta'}(x)]\big|_{\theta^\star,\theta^\star} \tag{21}$$

Recall the definition of $\pi_\gamma \mathcal{G}_\lambda^\infty(\theta)$ from (10). Since γ and λ are fixed, denote $\pi\mathcal{G}(\theta) = \pi_\gamma \mathcal{G}_\lambda^\infty(\theta)$. Finally, let $\mathcal{J}(\pi\mathcal{G}(\theta)) := \nabla_\theta \pi_\gamma \mathcal{G}_\lambda^\infty(\theta)|_\theta$ denote the Jacobian of $\pi_\gamma \mathcal{G}_\lambda^\infty(\theta)$.

I. There exists an open neighborhood $U \subseteq \Theta$ containing $\theta^\star$ such that for all $\theta \in U$, we have

$$\mathcal{J}(\pi\mathcal{G}(\theta)) = -\left(\nabla^2_{\theta',\theta'} \mathcal{H}(\theta, \pi\mathcal{G}(\theta))\right)^{-1} \cdot \lambda \nabla^2_{\theta,\theta'} \mathcal{H}_2(\theta, \pi\mathcal{G}(\theta)). \tag{22}$$

II. We have that $\nabla^2_{\theta,\theta'} \mathcal{H}_2(\theta^\star, \theta^\star) = \frac{B}{1+\gamma}$, and $B = -C$, so the Jacobian of $\pi\mathcal{G}$ at $\theta^\star$ is

$$\mathcal{J}(\pi\mathcal{G}(\theta^\star)) = (I + \lambda A^{-1}C)^{-1} \cdot \frac{\lambda}{1+\gamma} A^{-1}C \tag{23}$$

III. The spectral norm of $A^{-1}C$ can be bounded as

$$\|A^{-1}C\| \leq 1 + \frac{L\varepsilon}{\alpha}. \tag{24}$$

Proof. **We first prove I.** We apply Proposition A.5. Part A of that proposition gives us a function $g : U \to \mathbb{R}^d$ such that $\nabla_{\theta'} \mathcal{H}(\theta, \theta')_{\theta, g(\theta)} = 0$. But part B of that proposition says that there exists a unique local maximizer inside U, and this local maximizer is $\pi_\gamma \mathcal{G}_\lambda^\infty$. This implies that $\nabla_{\theta'} \mathcal{H}(\theta, \theta')_{\theta, \pi_\gamma \mathcal{G}_\lambda^\infty(\theta)} = 0$. Next, we implicitly differentiate this equation with respect to θ. Recall that when you have an equation of the form $f(x, y) = 0$, and implicitly differentiate it in the form $f(x, g(x)) = 0$ with respect to x, you obtain $\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\frac{\partial g}{\partial x} = 0$, and solving for $\frac{\partial g}{\partial x}$ yields $\frac{\partial g}{\partial x} = -\left(\frac{\partial f}{\partial y}\right)^{-1} \frac{\partial f}{\partial x}$. We apply this formula with

$$(x, f, g) = (\theta, \theta \mapsto \nabla_{\theta'} \mathcal{H}(\theta, \theta')_{\theta, \pi_\gamma \mathcal{G}_\lambda^\infty(\theta)}, \theta \mapsto \pi_\gamma \mathcal{G}_\lambda^\infty(\theta))$$

and obtain (22), as desired.